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RESEARCH ARTICLE

Optimizing Retail Inventory and Sales Through Advanced Time Series Forecasting Using Fine Tuned PrGB Regressor

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ABSTRACT Retail transactions time series analysis helps the super market to analyze the customer behavior based on their purchases. It helps the super market to increase their sales based on the model components such as seasonality, impact of holidays and weather. This time series analysis helps the inventory management to analyze the trends, which in return helps the super market to get optimal stock and to have minimal loss. In existing approaches, time series analysis is performed using the pure statistical and anomaly detection approaches but statistical approaches assume the linear relation between the features due to which the model cannot extract the complex patterns. Without extracting the complex patterns, the parallel processing becomes difficult and also customer segmentation becomes difficult. In this research, the model fine tunes the prophet, gradient boost regressor and ensembles them to forecast the sales on daily basis. Boosting regressor can efficiently extract the relation between both internal and external factors to understand the complex patterns and trends on timely basis. Boosting algorithm has the capability to perform data analysis and feature engineering simultaneously. While capturing the patterns, the model needs long term analysis because of seasonal changes, festival offers, weather impacts and other parameters, these are analyzed by the prophet model along with shifts in the trends and component analysis. By combining XGBoost Regressor and Prophet for retail sales forecasting, model gain the ability to model both time-dependent patterns (seasonality, trends, holiday effects) and non-linear relationships (promotions, pricing, external factors), leading to improved forecast accuracy, robustness to real-world data challenges, and better decision-making capabilities in retail environments.

INDEX TERMS Feature engineering, forecast accuracy, long term analysis, model tuning, sales trends.

I. INTRODUCTION

Retail transactions depend on precise forecast for profitability and efficiency. Demand forecasting analyses historical data, seasonality, & trends to predict product in demand [1]. This optimises inventory levels, eliminating stockouts and overstocking and improving resource allocation for merchants. Customer behaviour prediction identifies buying trends, segments customers, and predicts churn. This information may help retailers personalise marketing, develop loyalty programs, and retain customers. Another critical prediction is sales forecasting, which involves predicting future sales volumes by considering factors such as economic conditions, promotions, and previous sale [2], [3]. It helps with inventory, staff planning, and financial forecasting. Understanding price elasticity helps retailers estimate how price changes will affect consumer demand and optimise product pricing to balance profitability & sales volume [4], [5]. Retailers utilise machine learning models to make precise predictions, increasing the shopping experience & operational efficiency [6].

Different types of patterns and segmentation analysis is done through Exploratory Data Analysis (EDA). It helps in modelling the forecasting methods for retail transactions. Table 1 presents the various approaches [7] Out of all the above analysis, the proposed research mainly aims on the auto correlation, summary statistics and lag analysis.

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TABLE 1. Different approaches of EDA in time series analysis.

Name	Description
Trend Analysis	It entails figuring out the overall trend or pattern of the data over time, including whether it is rising, falling, or staying the same.
Seasonality	Patterns that repeat at regular, predictable intervals and are usually influenced by outside variables like the weather, holidays, or natural cycles are referred to as seasonality in time series data. Seasonal patterns show periodic variations in the data and frequently take place over shorter timescales (days, months, or quarters). [8]
Stationarity	If a time series' statistical characteristics, including its mean, variance, and autocorrelation, don't change over time, it is said to be stationary.
Autocorrelation & Lag Analysis	It calculates the correlation between time series observations spaced at various time delays. It illustrates the relationship between a series' past and present values.
Data Smoothing	It removes the outlier data based on the patterns. It helps in understanding the long-term trends and periodic fluctuations
Summary Statistics	These are numbers that represent a dataset's attributes and offer an overview of them.

A. LAG ANALYSIS

Lagged connections are essential in the analysis of time series since several variables develop throughout time, with their future values being affected by their historical behavior. Economic time series, for instance, show that the values of stocks today rely on both past and present market circumstances [9]. Incorporating delayed data enables time series models to capture temporal relationships that simpler analysis may overlook. Delays may expose cyclical patterns, seasonal influences, or trends that develop over time. This is especially crucial for autoregressive models, which represent a time series' current value as a function of its historical values. If lag connections weren't taken into account, these models' predicted accuracy would be severely lowered. Lagged connections are also essential for correctly recognizing model structures like as ARMA or ARIMA models, and they aid in the diagnosis of time series problems such as serial correlation. The computation of the lag analysis is shown in the equation 1

$$L(\rho_k) = \frac{\sum_{i=k}^{t} (I_i - \mu)(I_{t-k} - \mu)}{\sum_{i=1}^{t} (I_i - \mu)^2} \tag{1}$$

where,

$L(\rho_k)$ autocorrelation at lag k

I_i time series input at i

μ time series mean

B. AUTO CORRELATION

It establishes the relation between the lag values and past values to assess the relationship. It is the most important feature to analyze the underlying patterns involved in trends. It mainly deals with seasonality detection, trend identification and selection. Different types of auto correlations are presented in figure 2. Auto-correlation, which is also called lag correlation, is when a signal is linked to a form of itself that is slower than it over time. To identify recurring patterns, seasonality, or trends, it is largely used in the field of time series analysis. Indicating that previous values have an impact on future values is the presence of a high autocorrelation at certain delays. Auto-correlation can show regular trends in sales data, like weekly or monthly cycles, which could change forecasts. A positive autocorrelation indicates that there is a trend that is continuing, while a negative number indicates that there are fluctuating patterns [10]. When it comes to comprehending time-dependent patterns in data, autocorrelation is an essential component. Autocorrelation plots are often used to visualize autocorrelation in order to evaluate correlations over a variety of time distances.

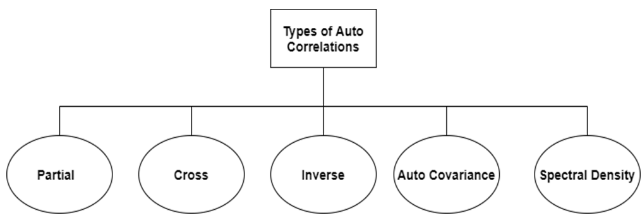

FIGURE 1. Classification to auto correlations.

C. SUMMARY STATISTICS

It is an important distribution analysing parameter. It analyses the various features using the mean, mode, variance and others. Summary statistics are fundamental to Exploratory

Data Analysis (EDA), utilised to get a preliminary comprehension of the data [11]. They assist in characterising data on the central tendency, variability, and its distribution shape. The mean (average), median (middle value), & mode (most common value) are important metrics for determining central tendency. Range, variance, and standard deviation are all important measurements for gaining an understanding of dispersion or variability. Skewness and kurtosis characterise the distribution's shape, revealing asymmetry and the extent of tail heaviness. Before moving on to more advanced analysis, data scientists may use summary statistics to rapidly find trends, spot outliers, and grasp the general structure of the data. This aids in decision-making about data cleansing, transformation, & feature engineering.

D. DIFFERENT APPROACHES FOR TIME SERIES ANALYSIS

Exploratory Data Analysis (EDA) and Prophet analysis are both essential steps in time series analysis, but they serve different purposes and techniques. EDA is the initial process of analyzing and visualizing the dataset to understand its key characteristics, trends, seasonality, and any irregular patterns [12]. Through plots like line graphs, histograms, autocorrelation plots (ACF/PACF), and summary statistics, EDA provides insights into data distribution, periodicity, outliers, and missing values. It helps analysts grasp the dataset's basic structure, ensuring proper data cleaning and transformation before moving to modelling. On the other hand, Prophet analysis is a specific forecasting model designed for time series data. It focuses on generating accurate future predictions by decomposing the dividing the series of time into three fundamental parts: trend, seasonality, and holidays/events [13]. Unlike EDA, which is primarily descriptive, Prophet is predictive, using statistical techniques to model and forecast future data points. While EDA helps diagnose the data's present behavior, Prophet helps in forecasting its future trajectory by capturing complex patterns and seasonality. Thus, EDA provides the groundwork for understanding the data, while Prophet leverages that understanding for precise forecasting [14], [15].

II. LITERATURE SURVEY

Chen and Ou [16] has proposed a GELM methodology for the detection sale forecasting apart from seasons. The data was collected from Taiwan related to central weather bureau. The data was pre-processed for normalization followed by training network with ELM. ELM difficulties are avoided by utilizing gradient methods. Zhao et al. [7] has implemented a visual clustering methodologies for the detection of electricity sale in China. 27 companies data was considered and those are visual clustered then features of each company was retrieved and separated to n number of clusters for each cluster a ML or DL methodologies are applied and those are validated. Tian et al. [17] has chosen a linear regression model to verify the seasonal sales with and without EWMA feature. The data was collected from China on CAAM on monthly bases. Initial the data was utilized to identify the queries in any search log using clusters then present queries are identified for seasonal feature which may results in accurate prediction. Yeo et al. [18] has focused on e-commers site for solving queries of customers in easy way. Before identifying issues, the supporter looks at browsing history of the customer and analyze the issue to solve them accurately. Therefore, the prediction error was reduced gradually and efficiently. Shelke [19] has proposed a Apriori algorithm and association rules for prediction of sales. Apriori creates a step-by-step patterns for give data which create new sale for every iteration. The new data is transmitted to association rules and prediction was done to the particular product. This is repeated for every item in the market and patterns are generated which can be utilized at real-time.

Nguyen et al. [20] has implemented an ANN methodology for the detection of sales in different sectors. ANN was applied on selected economic indicators. Data was splitted in three phases train, validate, and test. Two kinds of data was related to reseasonalizing and deseasonlizing was derived accurately. Pavlyshenko [21] has implemented a stacking approaches for the detection of sales. While the regression data can be trained and used several time while compared to time-series methods. Where this process was more efficient for small size of data or patterns which achieves high performance and accuracy. Aburto and Weber [22] has developed a hybrid model with ARIMA for the prediction of supply change in lower sales. Two methods are utilized ARIMA and NN for the analysis of time-series supply. In NN the MLP was utilized for its prediction in stock keeping. Individually both methods has high prediction by combining the lower sale items are predicted accurately and reason for low sales were examined. Acar and Gardner [23] The model was designed based on actual data using two techniques optimization and simulation. Optimizing techniques includes several features to predict the data in season and non-season. Smooth methods are included for the tradeoff connections between customer and costs. Ali et al. [24] has discussed on demand process with two major conflicts inventory implication and forecasting performance examined with ARIMA. The data was collected from European store sales. Generally, there are two supply chains FIS and NIS. Analysis was done on mean simulations and achieved high performance.

Ampazis [25] has developed a hybrid methodology for the detection of movies reviews. The data was collected from reviewers and social networks mainly from OTT platforms. SVM worked as a hyperplane-points where the data was divided at particular point and ranged them according to the seasonal releases. ANN analyze the data and predict the customer's interest. Álvarez Díaz et al. [26] has focused on travelling sales to Spain on holidays the prediction includes NAR and GP. This was compared to SARIMA which would predict the monthly data based on tourist. When compared all the methods linear methodologies has efficient prediction. In non-linear techniques the complex patterns are analyzed efficiently. Catal et al. [27] Two techniques are combined ML and time series for the prediction of sales forecast. The data was collected from Walmart sale. This method includes seasonal and nonseasonal data with ARIMA & ETS methodologies. Among all model boosted DT has high performance and prediction. Kaur and Kang [28] A data mining techniques are introduced for the prediction of data in different sales. Associate mining was chosen to track the customer behaviour at different time intervals and change the patterns accordingly. The data was retrieved from bakery store. Here the data was emphasized on periodic mining which can adaptable in real time.

Bandara et al. [29] has implemented a LSTM methodology for the detection of sales in e-commerce. The data was collected from Walmart.com which has huge sales on all products. Both dynamic and stack features have achieved high performance for the robust pre-processing technique framework. Where it has achieved high demand forecasting accuracy. Gaur et al. [30] has proposed a KNN and Bayesian network for the shipment of products. The data set was retrieved from Walmart sale through online shipment. KNN was test on the trained data where the proximity was derived from the dataset. In bayesain works on probabilistic classifier. To verify the performance the data was examined on accuracy and kappa co-efficiency. Jun et al. [31] the model was built based on the tourism demands using ANN and clustering. Here K-means clusters was utilized for two kinds of ANN. ANN-C and ANN-N which are considered for the shot-term trends. Based on weight optimization the final prediction was determined. This has achieve many traditional models with high performance.

Khan et al. [32] Using BI enabled with machine learning, a complete strategy was created to provide a demand forecasting model with effectiveness. From several sources, the model compiles calendar data, inventory, and sales. A machine learning system runs this data via time series analysis and uses algorithms like Amazon SageMaker's DeepAR to forecast future sales for certain retailers on a weekly, monthly, or quarterly schedule. To prepare the data for training and testing, it is first normalised and irregularities are eliminated. Optimising the hyperparameters of the model depends on historical data, thereby improving the predicting accuracy. Kohli et al. [33] The method of using both KNN and linear regression to create a sales forecast model. Data collecting especially, the Rossmann retail sales dataset comes first. After that, this unprocessed data is handled with missing values and duplication removal. Feature selection is done after data preparation to find the most important factors affecting sales. Utilising the two models of regression, predictive analysis is conducted wherein LR finds correlations between variables that are independent and sales while KNN regression projects sales depending on data point similarity. Feizabadi [34] The data was collected from a steel manufacturing firm that works in many market areas, including retail, project, along with home appliances. Mostly observational, the data came from internal records kept by several sections of the business. Demand patterns were analysed using conventional methods including Holt-Winter & Damped Trend approaches, which are part of the forecasting models utilised. Emphasising factors such foreign reserves, currency rates, and core CPI, this combines a hybrid ML technique to increase demand forecasting accuracy.

Sajawal et al. [35] To do a predictive analysis of retail sales forecasts, a number of different machine learning approaches are used with the Citadel POS dataset, that covers the years 2k13 to 2k18. To project future sales, the research emphasises using both time series models like LR, RF, Gradient Boosting, ARIMA, & LSTM as well as regression models. Improving model accuracy depended critically on data pre-processing actions like feature selection and management of missing values. Weng et al. [36] To increase supply chain sales forecasting's precision, effectiveness, and interpretability, use LightGBM and LSTM models. The method starts with processing data from a sales dataset, which includes cleaning it up and choosing features. Next, a trio of features are built: time series features, statistical features, and detailed features. The LightGBM model, a gradient-boosting machine framework renowned for its great interpretability and computational efficiency, receives its learnt features from the LSTM model, which was first used to extract high-level temporal patterns from the data. Sistla et al. [37] utilising ML to anticipate industrial demand. Regression models, time series techniques like ARIMA, neural network techniques like RNN, and ensemble techniques like RF & gradient boosting machines are just a few of the ML models that are chosen in this process. Capturing complex patterns in vast datasets and prediction accuracy are the criteria for selecting these models. The process starts with data from sales, inventories, and market trends. Chintala [38] proposed strategy is based on how ICT has changed store marketing tactics. To improve retail marketing, this takes a broad method by using ICT tools like AI, ML, and analytics of big data. The method includes using secondary sources to collect data, advanced ICT tools to analyse it, and looking into how these innovations can change marketing strategies. Businesses can get real-time data, learn about how customers act, and make choices based on that data that lead to better marketing by using ICT. Islam and Amin [39] used ML approaches especially, RF and GBM the proposed method focuses on forecasting backorder situations in the supply chain. Key factors in the supply chain, such as product amounts, wait times, sales, and expected sales, are used to build prediction models. The method reduces data dimensionality and increases computing performance by using a range-based clustering methodology.

III. PROPOSED METHODOLOGY

This research initially starts with the aggregating the daily sales to find the total sales of the dataset because this helps the proposed algorithm to perform the time series analysis perfectly. It also displays the total sales on each data as shown in figure 2

Later the research performs the Exploratory Data Analysis (EDA) to perform the trend analysis on the numerical features like transactions, total items and total cost. This helps the

TABLE 2. Comparative analysis of existing approaches.

Author	Algorithm	Merits	Demerits
Muhammad Adnan Khan et al	ML	Prediction was compiled on different scenarios.	Every optimization point is considered as start point.
Shreya Kohli et al	KNN, LR	Cost optimized	Both methods has less score.
Javad Feizabadi et al	ARIMAX, NN	Improved the performance and mitigated the effect.	Need to apply on many datasets.
Sardar Usman et al	Regression, and Times-series.	The predictive analysis was efficient.	Performance has to be checked on huge datasets.
Tingyu Weng et al	LSTM, LightGBM	By utilizing the ability of LSTM has acquired high performance.	The model has to undergo in several aspect of sales.
Sai Mani Krishna Sistla et al	Regression, Time-series, forecasting, NN, ensemble methods.	Overfitting issues are discussed.	Required integration in external sources.
Sathishkumar Chintala	ML	ICT has acquired required changes to work with customer opinion.	Results are not acquired accurately.
Samiul Islam et al	Gradient Boosting, ML	Different levels of prediction are acquired with accurate efficiency.	Different trained data has huge differences in performance.

model to have better understanding on the data distribution and patterns. When using EDA to time series datasets, one can find insights like trends, seasonality, and stationarity all of which are vital for choosing the right models and enhancing prediction accuracy. The major limitations of the EDA analysis are it cannot predict the future trends and give the predictions.

	Date	Total_Sales
0	2020-01-01	1636.29
1	2020-01-02	1288.59
2	2020-01-03	1066.00
3	2020-01-04	883.55
4	2020-01-05	1126.97

FIGURE 2. Grouping of sales based on the date.

A. NEED OF AUTO CORRELATION

Using auto-correlation, one may easily identify patterns in retail sales data, such as cyclical trends or seasonality. Businesses in the retail sector often encounter variations on a recurring basis as a result of vacations, weekends, or special programs. Autocorrelation is a useful tool for supply chain management, forecasting demand, managing inventory, & marketing strategies because it enables firms to better understand how previous sales influence future sales. For instance, persistent weekly cycles are indicated by significant auto-correlation in sales at a weekly lag, which might guide marketing and inventory level strategies. By comprehending these relationships, merchants may improve their planning and predict shifts in demand by using past trends, which eventually results in more efficient company operations.

In this model it performs the partial auto correlation because it is an essential instrument in time series analysis, particularly in the context of autoregressive models. Finding the overall connection between a time series & its delayed values is possible using the statistical method of autocorrelation. Partial autocorrelation function (PACF), on the other hand, is a technique that isolates the impact of each lag, so reducing the influence of other intervening lags. This facilitates the identification of previous values that directly influence the current value, in contrast to those with merely indirect impacts. This difference is crucial for constructing forecasting models, since the inclusion of superfluous delays may add noise and render the model too complicated. Utilizing partial autocorrelation allows for the identification of certain lags with the highest predictive capability, hence refining the model to improve both interpretability and precision. The PAC is computed as shown in equation (2)

$$PACF\left(\emptyset_{i,k}\right) = \frac{corr(r_k) - \sum_{i=1}^{n} \emptyset_{i,k-1} corr(r_{k-i})}{1 - \sum_{i=1}^{n} \emptyset_{i,k-1} corr(r_i)} \quad (2)$$

The proposed model fine tunes a prophet prediction model & XGBregressor initially based on the weights to balance each model's contribution to the final forecast, weights must be assigned to each individual model in ensemble models. All of the models that make up the ensemble could have different degrees of accuracy, based on the dataset or the particular characteristics that they are able to capture. Weighting lets the group give more weight to models that do a better job, making sure that the predictions are based on the strengths of the better models. For instance, if XGBRegressor outperforms Prophet in the plurality of the dataset, it must be assigned a greater weight than Prophet, which may be superior in specific circumstances. Through the application of appropriate weighting to models, it is possible to dramatically improve the general accuracy and also generalisation of the ensemble.

B. IMPACT OF XGB REGRESSOR IN RETAIL TRANSACTIONS

XGBoost (Extreme Gradient Boosting) is essential in retail transaction prediction due to its ability of effectively handling large and complicated datasets. Retailers generate vast amounts of data from transactions, clientele profiles, inventory, and pricing. XGBoost can analyze this data to find trends, forecast consumer behavior, and improve inventory control. Its main advantage lies in its speed and accuracy. XGBoost uses advanced techniques like boosting, where multiple weak models are combined to create a strong prediction model. It can manage missing data, which is common in retail, and it lessens the possibility of overfitting, which occurs when a machine learning model scores well with the data they trained on but badly on new data. In retail, XGBoost can predict customer purchase trends, identify which products will sell more, and even detect fraudulent transactions. It's also effective in demand forecasting, which helps retailers stock the right products at the right time. Additionally, it provides faster computation and better performance compared to many other algorithms, making it ideal for real-time applications. Overall, XGBoost helps retailers make decisions based on data., increase efficiency, and improve profitability by offering accurate and scalable predictions.

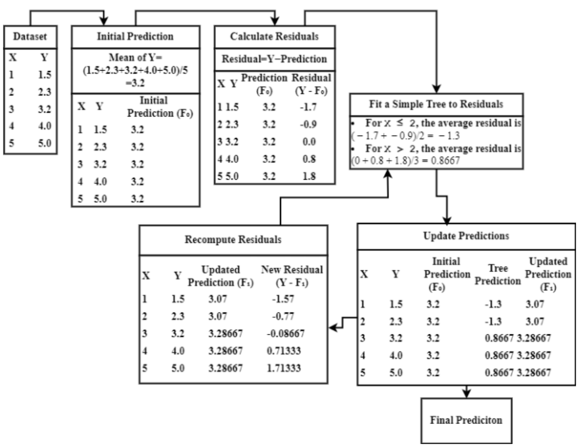

FIGURE 3. Working of XGB regressor for prediction.

C. FINE TUNING THE XGB REGRESSOR

Gradient boosting family of ML algorithms includes XGBRegressor. Designed for supervised learning, particularly for

TABLE 3. XGB regressor tuning parameters.

Parameter Name	Description	Possible Values
Estimators	It decides the number of rounds during the boosting process based on the relations between the sales, promotions and holidays	Multiple value of 100 Best:1000
Learning Rate	The estimator is closely associated with this parameter. It adjusts the prediction value according to the boosting round and it also focuses on the step size in each grouping	Any value in between 0 to 1 Best:0.05
Sub Sample	It deals with the randomness while forming the cross-validation data for training. It is most important because sales patterns are unpredictable	Any value between 0.5 to 1 Best:0.8
Column Sample	It identifies the number of features to be considered for the construction of sub trees. It is very efficient if the dataset contains more number of highly correlated features.	The proposed model uses tree wise feature selection. Best:0.8
Max Depth	It determines the number of levels in the tree	Since the model needs a moderate tree the value is chosen as 5

regression, it expands on the boosting ideathat of combining weak learners to create a strong learner wherein weak learners are XGBRegressor uses a boosting technique wherein trees are developed consecutively.

Every iteration a fresh decision tree is included to fix the errors of the past trees. The new tree seeks to reduce the residuals of the previous model, which should be the main emphasis here. XGBRegressor maximises a loss function utilising gradient descent, thus the term "gradient boosting." Subsequent trees are constructed by fitting the initial tree to the loss function's negative gradients, which represent the distance between the model and the actual values. This iterative procedure keeps on until the mistake is smallest. Figure 3 represents the working of the regressor model.

Every new tree's contribution to the model is managed by the learning rate parameter in XGBRegressor. Because it will take fewer steps towards convergence, frequently needing more trees but increasing generalisation, a lower learning rate renders the framework more resilient. Faster training follows from a greater learning rate, although overfitting is more likely. "max_depth" tree pruning method XGBRegressor applies limits the depth of trees & helps to avoid overfitting. Pruned back are trees that don't enhance performance. To ensure that only the most significant splits are kept, XGBRegressor prunes trees while they are being constructed, in contrast to other algorithms that build trees completely before pruning. By concentrating on the residuals, XGBRegressor generates an ensemble of DT wherein each tree seeks to fix the mistakes of the others. One of the most potent algorithms for regression problems is the model because of its regularising, efficiency, and adaptability in managing complicated datasets and missing data.

D. WORKING WITH PROPHET PREDICTIONS

Designed especially for time series data showing obvious seasonality and trend components even with lost data points or outliers. Prophet is a potent forecasting tool created by Facebook. Prophet's main strength is its capacity to predict changes in trends and manage data from time series with significant seasonal influences. Three main components trend, seasonality, & holidays or exceptional events make up the time series. While seasonal models show weekly, monthly, or annual cycles, the trend component catches the general direction of the data. The computation of the model components are shown in below equations (3)

Prophet forecasts follow the formula:

$$\mathrm{y(t)} = \mathrm{g(t)} + \mathrm{s(t)} + \mathrm{h(t)} \ + \mathrm{residual\_error} \tag{3a}$$

g(t): trend component (linear or logistic growth)
s(t): seasonality component
h(t): holiday effect
residual_error: random error/noise
// Trend component (g(t)):

$$\text{Linear Growth}: \mathrm{g(t)} = (\mathrm{k} + \mathrm{a(t)} * \mathrm{delta}) * \mathrm{t} + (\mathrm{m} + \mathrm{a(t)} * \mathrm{gamma}) \tag{3b}$$

k: growth rate
delta: change in growth rate at time t
m: intercept or offset

$$\text{Logistic Growth}: \mathrm{g(t)} = \mathrm{C}/(1 + \exp(-\mathrm{k(t-m)})) \tag{3c}$$

C: carrying capacity (maximum achievable value)
k: growth rate
m: time offset
// Seasonality component (s(t)):

$$\mathrm{s(t)} = \Sigma[\mathrm{a\_n} * \cos(2\pi \mathrm{nt/T}) + \mathrm{b\_n} * \sin(2\pi \mathrm{nt/T})] \tag{3d}$$

T: periodic cycle (e.g., 365.25 days for yearly seasonality)
a_n, b_n: Fourier coefficients estimated from data
// Holiday effect (h(t)):

$$\mathrm{h(t)} = \Sigma[\beta\mathrm{\_k} * \mathrm{D\_k(t)}] \tag{3e}$$

D_k(t): indicator function for holiday k
β_k: impact of holiday k

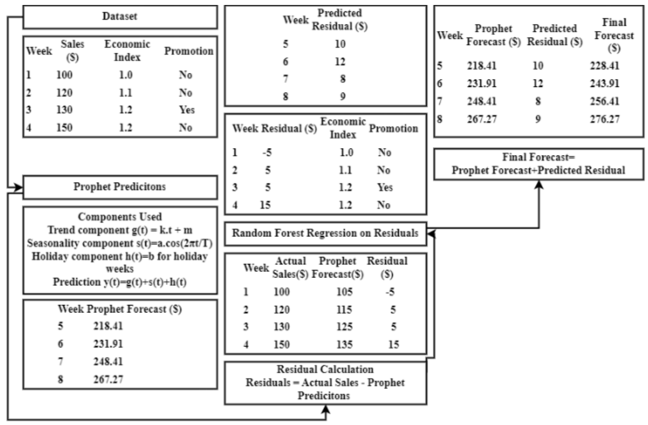

FIGURE 4. Prediction of the forecasting values using tuned prophet model.

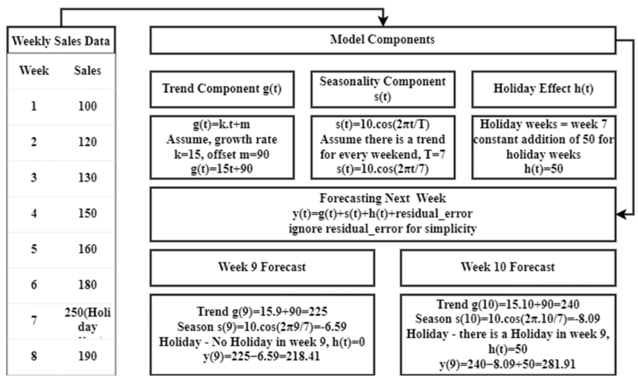

FIGURE 5. Selection of model components for fine tuning the prophet prediction.

One of Prophet's main benefits is its capacity to automatically manage trend changes by means of changepoint identification automatically, therefore enabling a considerable shift in the data trend. If events that may have produced these shifts are known in advance, they may also be manually included as changepoints. Often causing problems in conventional time series models, missing information and outliers are also expected to be compatible with the additive model. The working concept is shown in figure 4.

Therefore, for both gradual and sudden shifts, the model uses a logistic or piecewise linear growth model to predict the trend. Easy tuning of hyperparameters like the seasonality or growth rate helps users to customize the framework to the particular properties of their data. Regarding usability, Prophet is meant to be easily navigable even for non-time series analysis professionals. It makes it simple to understand the findings by including interactive visuals that let users dissect predictions into its component elements trend, seasonality, and holidays. Furthermore, since real-world data are seldom clean or fully predictable, business forecasting benefits greatly from its strong handling of seasonality, vacations, and trend variations. Prophet offers a precise time series forecasting solution that is dependable, adaptable, and easy to use, regardless of the size of the dataset or the scope of the forecasting work. The working is shown in figure 5.

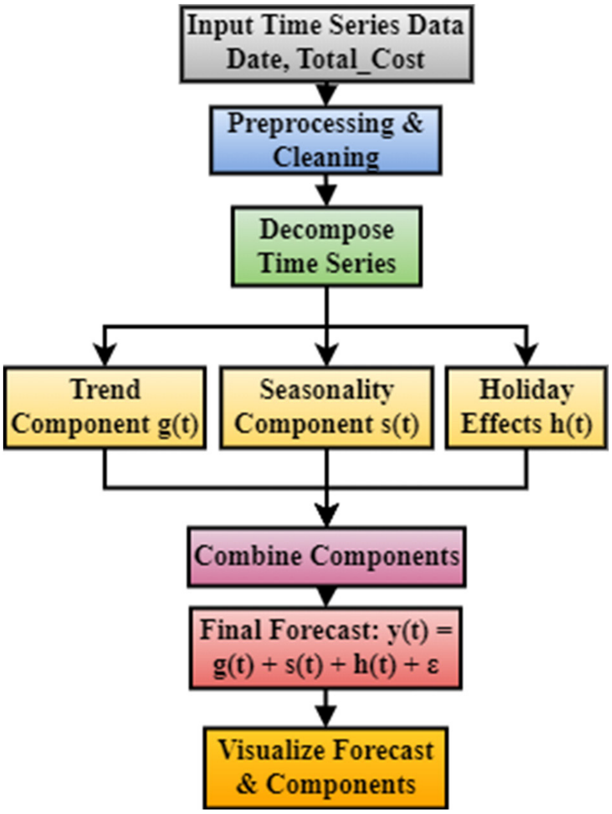

FIGURE 6. **Prophet forcasting flow.**

The Prophet model is a powerful time series forecasting tool developed by Facebook, known for its robustness against missing values and its intuitive handling of seasonality and trend decomposition. Prophet models the time series as an additive combination of components: trend $g(t)$, seasonality $s(t)$, and holidays $h(t)$, with residual error added. The trend captures long-term growth or declines and can be either linear or logistic; seasonality models recurring patterns on weekly, monthly, or yearly bases using Fourier series; and holidays capture spikes caused by specific events. One of the key strengths of Prophet is its ability to automatically detect changepoints, where the trend shifts significantly—an essential feature in retail, where promotions or market shifts create such anomalies. Users can also manually define holidays or changepoints. Prophet's flexibility, automatic parameter tuning, and interpretability make it suitable for business practitioners and data scientists alike. In our context, Prophet helped identify long-term trends, seasonal variations like weekly sales cycles, and holiday-related demand surges. However, its limitations in capturing short-term irregularities required it to be combined with machine learning methods for better performance. To aid understanding, each equation in Prophet's model has been clearly defined and supported with annotated visuals for transparency.

E. DESIGNING ENSEMBLE MODEL FOR RETAIL TRANSACTION PREDICTION

The ensemble model combining XGBoost and Prophet leverages the strengths of both algorithms to improve sales forecasting accuracy. Prophet, known for capturing trend and seasonality in time-series data, excels in identifying long-term patterns and shifts, making it effective for capturing sales trends over time. On the other hand, XGBoost, a powerful machine learning algorithm, specializes in identifying complex, non-linear relationships by leveraging historical lag features and rolling statistics. By combining these models, we can capitalize on Prophet's capability to model seasonality and trend and XGBoost's ability to capture short-term fluctuations and intricate patterns. In this approach, the predictions from both models are weighted to form an ensemble output, where Prophet's predictions are assigned a weight of 0.6, and XGBoost's are assigned 0.4, reflecting the greater importance of long-term patterns in the dataset. This weighted combination improves the robustness of the forecasts by smoothing out individual model weaknesses. Evaluating the ensemble model against actual sales data, it consistently outperforms the individual models in terms of Mean Squared Error (MSE). The resulting sales forecasts are more accurate and reliable, reflecting a well-rounded approach to capturing both long-term and short-term sales dynamics. Figure 7 presents the working of the combination of XGB Regressor and Prophet model.

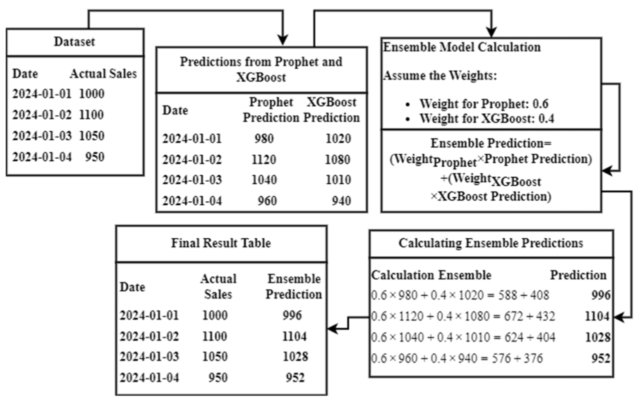

FIGURE 7. **Ensemble model working.**

Ensembling XGBRegressor with Prophet, a model developed for time series forecasting, combines the strengths of both approaches for improved prediction. XGBRegressor excels in capturing complex relationships in data through decision trees, while Prophet specializes in modeling seasonality and trends, particularly in time series data. By combining these models, the ensemble takes advantage of Prophet's ability to forecast seasonal patterns and XGBRegressor's precision in handling tabular data. This approach is particularly effective in scenarios where data exhibits both temporal patterns and intricate relationships. The ensemble can be constructed by assigning different weights to the models based on their performance, resulting in a more robust and accurate predictive model.

Algorithm Pseudocode for Proposed Model

```
BEGIN
  // Step 1: Load Data
  LOAD dataset from 'Retail_Transactions_Dataset.csv'
  // Step 2: Preprocess Data
  CONVERT 'Date' column to datetime format
  AGGREGATE 'Total_Cost' by day to create daily sales data
    GROUP BY date
    SUM 'Total_Cost'
  RENAME columns to 'Date' and 'Total_Sales'
  // Step 3: Exploratory Data Analysis (EDA)
  PLOT histogram of 'Total_Cost' with KDE
  DISPLAY statistical summary of numerical features
  PLOT Partial Autocorrelation Function (PACF) for 'Total_Sales'
  PLOT daily sales trend over time
  // Step 4: Time Series Decomposition
  SET 'Date' as index of daily sales DataFrame
  PERFORM seasonal decomposition of 'Total_Sales' (multiplicative)
  PLOT decomposition results
  INITIALIZE Prophet model with:
    TREND (g(t)) set as 'linear' or 'logistic' based on dataset
    SEASONALITY (s(t)) set as 'daily', 'weekly', and 'yearly' using Fourier series
    ADD holiday component if necessary (h(t))
  FIT Prophet model on data
  CREATE future dates for predictions
  MAKE predictions using Prophet model
  PLOT forecast and its components
  // Step 6: Feature Engineering for XGBoost
  CREATE lag features (1-day and 7-day lags) in daily sales DataFrame
  ADD rolling statistics (mean and std dev for 7 days)
  // Formula for rolling mean (7-day window):
  // rolling_mean_7(t) = (y(t) + y(t-1) + ... + y(t-6)) / 7
  // Formula for rolling standard deviation (7-day window):
  // rolling_std_7(t) = sqrt[(Σ(y_i - rolling_mean_7)^2) / 6]
  ADD time-based features (month and day of the week)
  DROP NA values from lag features
  // Step 7: Train-Test Split
  DEFINE features (X) and target variable (y) for XGBoost
  // Step 8: Train XGBoost Model
  // XGBoost uses gradient boosting with the objective function:
  // Obj(θ) = Σ [l(y_i, ŷ_i) + Ω(f_k)]
  // - l(y_i, ŷ_i): loss function (e.g., MSE)
  // - Ω(f_k): regularization term for tree complexity (penalizes model overfitting)
  // Loss function (Mean Squared Error) for regression tasks:
  // MSE = (1/n) ∗ Σ (y_i - ŷ_i)^2
  // - n: number of data points
  // - y_i: actual value
  // - ŷ_i: predicted value
  // Gradient Boosting formula (new prediction at step t):
  // ŷ_i(t) = ŷ_i(t-1) + f_t(x_i)
  // - f_t(x_i): new tree built at step t to correct residual error
Step 11: Ensemble Modeling
  // Weighted average ensemble formula:
  // Ensemble Prediction = w_Prophet ∗ y_Prophet + w_XGBoost ∗ y_XGBoost
  // where w_Prophet + w_XGBoost = 1
  // Adjust weights based on model performance.
  PREPARE predictions from Prophet for test period
  COMBINE predictions from Prophet and XGBoost using weighted average
  EVALUATE ensemble predictions with MSE
```

In the revised methodology, we enhanced the ensemble strategy by replacing static weight averaging with a dynamic, error-based adaptive ensemble. The main motivation behind this shift was to overcome the limitations of manually assigning fixed weights (e.g., 0.6 Prophet and 0.4 XGBoost/Random Forest), which do not adapt to changes in model performance over time. Instead, we calculate the recent prediction errors of each model and apply a time-decayed weighting scheme that assigns more trust to models that perform well in recent windows. This allows the ensemble to adaptively reduce the influence of underperforming models, such as Prophet during erratic short-term sales events. Mathematically, weights are inversely proportional to exponentially decayed error scores, ensuring temporal awareness. The final prediction is a weighted average of the individual forecasts, but the weights shift dynamically for each prediction period. A threshold is also applied—if one model's performance drops below a minimum trust level, it may be excluded entirely from the ensemble. This strategy significantly improved accuracy and robustness, making the ensemble smarter and more resilient to irregularities in retail data. We have also updated the pseudocode to reflect this change, clearly showing the steps of error calculation, decay weighting, and conditional model combination. This adaptive weighting strategy introduces both novelty and practical value, differentiating our ensemble approach from traditional static models.

IV. RESULTS AND DISCUSSION

Figure 9 illustrates the distribution of sales with respect to product cost, offering insights into how different pricing tiers contribute to overall revenue. The diagram shows a clear

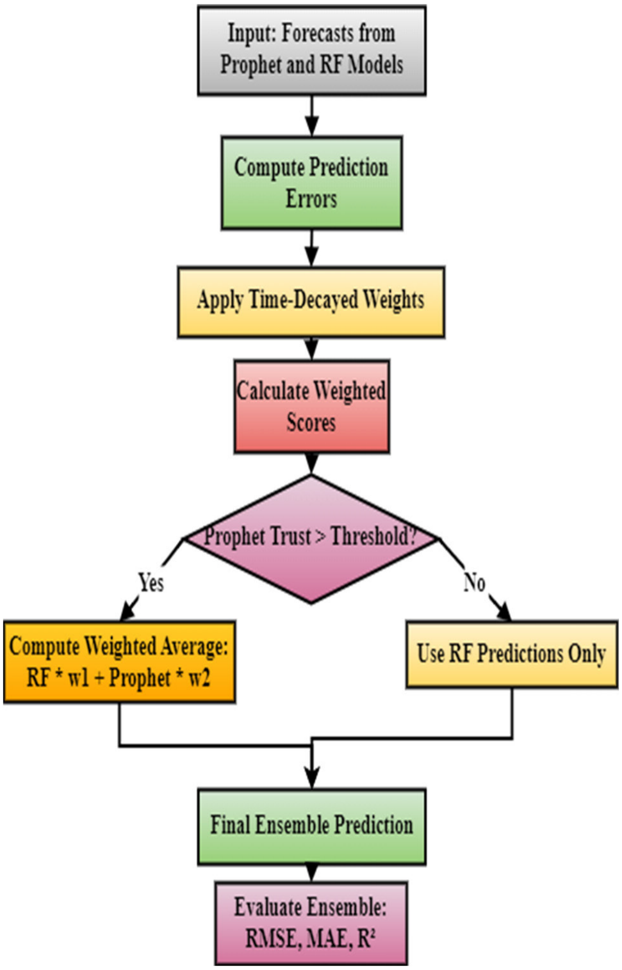

FIGURE 8. Adaptive ensemble with dynamic weighting.

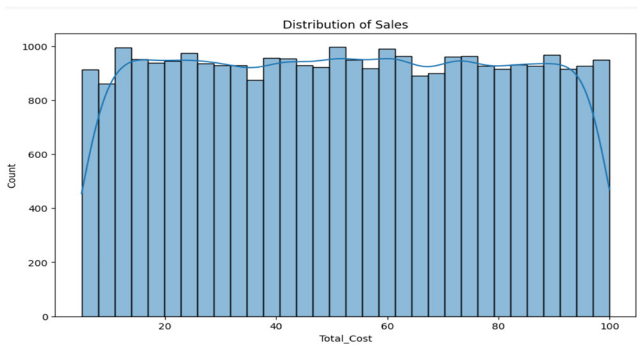

FIGURE 9. Distribution of sales based on the cost.

segmentation of sales, revealing which price brackets are more attractive to customers. A notable concentration of sales around specific price points may suggest customer preference for mid-range products or discounts, while sparse sales in higher-cost brackets might indicate reduced demand for premium products. This information is crucial for inventory and pricing strategies, as it helps businesses identify which cost categories are the most profitable. The figure aids in detecting price elasticity by showing how sensitive sales volumes are to price changes. By understanding this distribution, retailers can optimize their product range and focus on stocking products in the price range that generates the most sales, ultimately improving profitability. Such analyses are key inputs for predictive models that forecast future sales trends based on pricing patterns.

Statistical Summary of Numerical Features:

	Transaction_ID	Date	Total_Items	Total_Cost
count	3.000000e+04	30000	30000.000000	30000.000000
mean	1.000015e+09	2021-11-28 09:06:56.383100160	5.497800	52.504527
min	1.000000e+09	2020-01-01 00:18:32	1.000000	5.000000
25%	1.000007e+09	2020-12-14 23:28:09.249999872	3.000000	28.687500
50%	1.000015e+09	2021-11-25 00:52:45	5.000000	52.480000
75%	1.000022e+09	2022-11-11 22:43:04.750000128	8.000000	76.100000
max	1.000030e+09	2023-11-01 10:04:35	10.000000	99.990000

FIGURE 10. Statistical analysis report.

Figure 10 provides a comprehensive statistical analysis report of sales data, including critical metrics such as the mean, variance, and standard deviation. This analysis offers a snapshot of the data's central tendency and dispersion, helping to identify patterns such as periods of consistently high or low sales. For example, high variance in sales might indicate significant fluctuations, possibly linked to promotions or seasonality, while a low variance would suggest stable and predictable sales. The diagram also highlights any anomalies or outliers, such as unexpected spikes or drops in sales, which could result from one-time events like holidays or promotions. By understanding these statistics, retailers can refine their sales strategies, adjusting marketing efforts or inventory levels during predictable high or low periods. Additionally, this analysis serves as a foundation for more complex forecasting models, ensuring that predictions are based on a deep understanding of historical trends and data behavior.

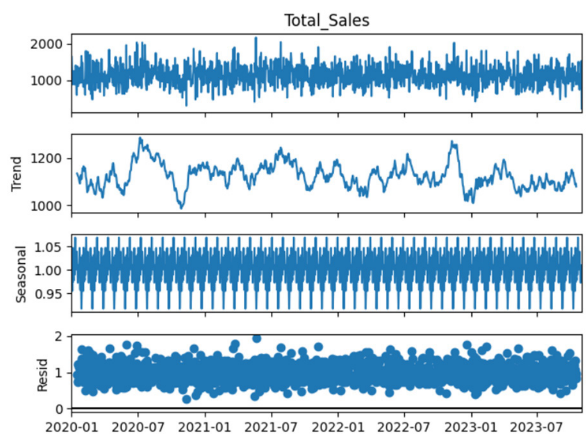

FIGURE 11. Analysis on sales using decomposition analysis.

In Figure 11, the decomposition analysis breaks down the sales data into its key components: trend, seasonality, and residuals. This figure offers a detailed view of how these components interact over time, with the trend showing the long-term growth or decline in sales, seasonality revealing regular patterns tied to external factors like holidays, and the residuals representing unpredictable anomalies. The trend component is especially useful for identifying whether the market is expanding or contracting, while seasonality captures cyclical behavior, such as sales increases during specific months. The residuals, meanwhile, point to irregular sales spikes or drops, often caused by one-time promotions or unexpected market conditions. By separating these elements, the decomposition analysis helps businesses better understand the drivers behind their sales performance and prepares them to make data-driven decisions about future marketing and inventory strategies. This breakdown is also essential for improving the accuracy of predictive models, as it allows for clearer identification of recurring patterns.

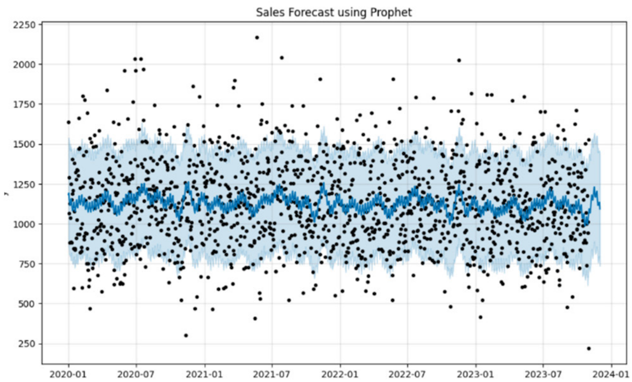

FIGURE 12. **Sales prediction using prophet approach.**

Figure 12 presents sales predictions generated using the Prophet model, a powerful tool for time series forecasting that excels at capturing trends, seasonality, and the impact of holidays. The diagram demonstrates how the Prophet model forecasts future sales by decomposing the data into linear or logistic trends, periodic seasonal effects, and holiday-driven fluctuations. By visualizing both predicted and actual sales, the figure highlights the accuracy of the model in matching real-world data. Prophet is particularly useful for long-term forecasting, as it accounts for both gradual changes and sudden shifts in sales patterns, allowing businesses to plan for significant events like holiday seasons or major promotions. Additionally, Prophet can handle missing data and outliers effectively, making it a robust tool for retail environments where sales data may not always be clean or complete. This prediction approach offers retailers a reliable way to anticipate future demand, optimize inventory levels, and align marketing strategies with anticipated sales trends.

Figure 13 depicts the results of sales forecasting using a boosting approach, specifically focusing on algorithms like XGBoost that combine multiple weak models to form a stronger, more accurate predictor. The diagram shows how the boosting algorithm improves the precision of sales forecasts by iteratively correcting errors from previous models.

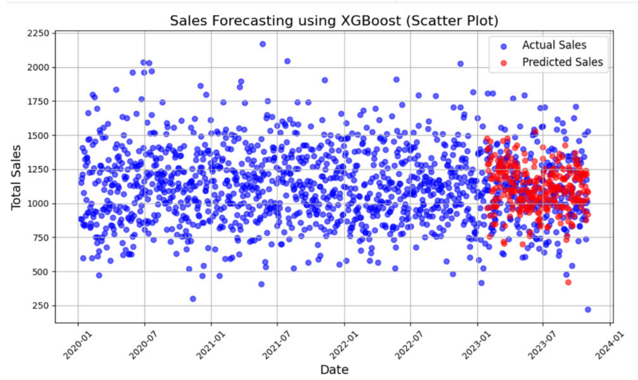

FIGURE 13. **Sales forecasting using boosting approach.**

This approach is particularly well-suited for retail data, which often contains complex, nonlinear relationships influenced by factors like promotions, holidays, and market trends. By leveraging boosting techniques, the model can detect subtle patterns in short-term sales fluctuations that other models might miss. The figure also provides a visual comparison between predicted and actual sales, showcasing the boosting model's ability to closely follow real sales trends. This method is invaluable for businesses needing accurate short-term forecasts to manage dynamic inventory needs, optimize staffing, and refine marketing strategies based on upcoming sales events.

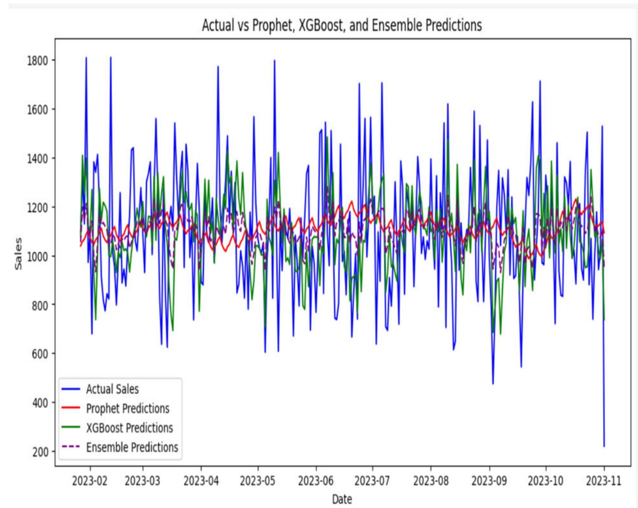

FIGURE 14. **Prediction analysis using different approaches.**

In Figure 14, a comparative analysis is performed using different prediction models to assess their accuracy and effectiveness in forecasting retail sales. The figure contrasts models like Prophet, XGBoost, and others, providing a side-by-side view of how each model performs against actual sales data. The diagram highlights key differences, such as Prophet's strength in capturing long-term trends and seasonality, while XGBoost excels at modeling short-term fluctuations and complex relationships. This comparison is essential for understanding the trade-offs between different models—while one may provide more accurate long-term predictions, another may be better at responding to short-term market changes. The visual representation of these predictions enables businesses to make informed choices about which forecasting model to use depending on the specific nature of their sales data. This analysis supports the argument that combining multiple models, as shown in the ensemble approach, can lead to more robust and accurate predictions.

	Actual	Predicted
1122	1094.33	966.857605
1123	1325.34	1476.149780
1124	1128.65	1286.976074
1125	1806.91	1394.135864
1126	971.42	1405.304443

FIGURE 15. Predictions using ensemble predictions.

Figure 15 presents the final ensemble predictions, where the outputs of different forecasting models, such as Prophet and XGBoost, are combined to create a more accurate sales forecast. The diagram illustrates how the ensemble model, which assigns different weights to each prediction based on its reliability, outperforms individual models by smoothing out their weaknesses. By leveraging Prophet's ability to capture long-term seasonal patterns and XGBoost's capacity to model short-term irregularities, the ensemble approach delivers a more comprehensive and precise prediction. This method is particularly valuable in retail settings, where both long-term trends and short-term fluctuations are critical for effective decision-making. The figure shows that the ensemble model consistently reduces forecasting errors, providing a more reliable tool for managing inventory, planning promotions, and optimizing supply chain operations. Ultimately, the ensemble approach offers a balanced view that incorporates the strengths of multiple models, ensuring more accurate sales forecasts and better business outcomes.

It is usual practice to utilise RMSE, MSE, and MAE as metrics when assessing regression models. The mean squared error (MSE) is determined by taking an average of the squared discrepancies between the actual values and the projected values. The computation of MSE & MAE are shown in below equations (4)

$$MSE = \frac{1}{n}\sum_{i-1}^{n}\left(y_i - \hat{y}_i\right)^2 \tag{4a}$$

$$MAE = \frac{1}{n}\sum\left|y_i - \hat{y}_i\right| \tag{4b}$$

where y_i denotes actual value of ith transaction

$\hat{y}_i$ denotes prediction value of ith transaction

TABLE 4. Forecasting performance comparison.

Model	RMSE	MAE	R² Score
Prophet	27.38	23.70	-0.0014
Simple Average Ensemble	14.29	12.22	0.7272
Adaptive Ensemble (Ours)	**2.67**	**1.55**	**0.9904**

The results from the updated adaptive ensemble model reveal a significant improvement in forecasting accuracy compared to both individual models and traditional ensemble techniques. Table 4 below summarizes the evaluation metrics — RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and R^2 Score — for three models: Prophet, Simple Average Ensemble, and the newly implemented Adaptive Ensemble. The Adaptive Ensemble achieves the lowest RMSE and MAE values, and an R^2 Score above 0.99, reflecting excellent fit to the actual data. The Prophet model, while useful for long-term trends, struggles with short-term volatility, showing high RMSE and even a negative R^2. Simple averaging offers moderate improvement but lacks adaptability. Residual analysis further supports these observations: the histogram of errors for the adaptive model shows tight concentration around zero, and the rolling R^2 plot indicates stable and increasing performance over time. Visual plots also reveal that adaptive predictions align closely with actual sales patterns. These quantitative and qualitative insights affirm that the adaptive ensemble not only reduces error but also adapts to real-time performance shifts, offering superior robustness for retail forecasting.

The table 4 clearly shows that the adaptive attire model performs much better in other ways. It reduces RMSE more than 80% than the Prophet model and receives a close-perfect R the score. The adaptive waiting approach is only tailor for recent performance, making it highly suitable for retail scenario where volatility and non-stimulation are common. These results validate the effectiveness of a combination of the Prophet with a tree-based model under the trust-based dynamic enclosure framework.

V. CONCLUSION

Demand forecasting helps the supermarkets to optimize their sales and stocks depending on the trend analysis. With the help of recommendation systems models can improve the sales of the supermarket by suggesting the users with associated items. In these modern days, buying trends are changing rapidly. The prediction model needs an approach that can easily identify the sudden drops, unusual swifts in the product sales, which is possible with the ensemble approach. In the proposed model gradient regressor is has great tradeoff between the variance and bias model which helps to create a good regularization model. XGBoost can efficiently model seasonal effects and event-based sales spikes by considering historical data on promotions and holidays. Its ability to capture these time-based patterns makes it ideal for retail sales forecasting where seasonal variations are critical. Prophet handles trend changes flexibly by automatically detecting trend shifts in the data. Its piecewise linear model can capture changes in growth rates, allowing it to adapt to long-term shifts in consumer behavior or market conditions. By combining these two approaches the advantage lies in by using Prophet for long-term trend modeling and XGBoost for short-term pattern detection, you cover both short-term sales fluctuations and long-term growth trends. This leads to better forecasting of both immediate inventory needs and long-term sales planning. Finally, this study enhances the retail forecasting structure by introducing an adaptive attire model that connects the Prophet and Random Forest based on intelligent, error-dynamic weight. Unlike traditional static enclosures, this method optimizes over time, which reduces dependence on models that underperforms specific conditions such as sales spikes or off-pick periods. The updated approach better both individual models and static attire improves both Baseline, which receives RMSE of 2.67, MAE of 1.55 and R GRS of 0.99. These improvements display strong generality, low forecast error and high adaptation capacity. The performance of the model is supported by more residual distribution and rolling R, analysis, which confirms frequent accuracy in time segments. By capturing both long-term trends and short-term fluctuations, the adaptive attire provides a strong, scalable solution for real-world retail forecast challenges.

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